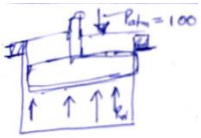
TYPICAL UNIT 1 CALCULATION MCQ'S IN ME 265 - DETAILED SOLUTIONS

1. Balancing vertical forces, we write: $P_w A = Mg + P_{atm} A$

$$P_w = \frac{Mg}{A} + P_{atm} = \frac{100 \times 9.81}{0.01} + 100{,}000 = 198100 \text{ Pa} = \mathbf{198.1\ kPa}$$



2. $v = 1.51.5 \frac{m^3}{kg}$ is $> v_g$ at 200 ℃,

hence its superheated vapour $1.08034 < 1.5\ m^3/kg < 2.17726$

hence, $P_{required} = 100 + (200 - 100)\frac{(1.5 - 2.17226)}{(1.08034 - 2.17226)} = \mathbf{161.6\ kPa}$

3. Energy balance for steam flow results in $q + h_{ground} = h_{top} + gZ_{top}$

$q = -100$ kJ/kg; $\quad h_{top} = h_{g@125\ kPa} = 2685.35$ kJ/kg

$h_{ground} = 2685.35 + \frac{9.81 \times 100}{1000} - (-100) = 2796.33$ kPa

$T_{ground} = T_{sat@2796.33\ kPa} = \mathbf{163.54}$ ℃

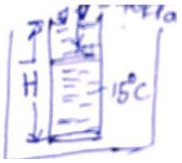
4. Initial specific volume $= v_1 = 0.5\ m^3/kg$

Interpolating for the pressure of the same specific volume of $0.5\ m^3/kg$

$$= 400 + \frac{(0.53422 - 0.5)}{(0.53422 - 0.42492)}(500 - 400) = \mathbf{431.31\ kPa}$$

5. $m = \rho Volume = \frac{Volume}{v} = \frac{AH}{v}$; $\quad v_{@15℃} = 0.001001\ m^3/kg$

$$m = \frac{200 \times 10^{-4} m^2 \times 5m}{0.001001\ m^3/kg} = \mathbf{99.9\ kg}$$



6. $\frac{P_i V_i}{T_i} = mR = \frac{P_f V_f}{T_f}$

$$T_f = \frac{P_f V_f T_i}{P_i V_i} = \frac{350 \times 800 \times 10^{-3}}{150 \times 400 \times 10^{-3}} \times 300\ K = \mathbf{1400\ K}$$

7. $\Delta u = c_v \Delta T = 0.657 \times (1200 - 600) = \mathbf{394.2\ kJ/kg}$